

POWERACT

CONSULTING

GAP ANALYSIS

SRS Gap Analysis

Dynamic Apps Standard SRS versus the CortexPLM application and its documents

Version 1.0 · September 2026

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1. Summary

CortexPLM meets **148 of 155** requirements of the Dynamic Apps Standard SRS, up from 138 in the first delivery. This review found 10 requirements that were only partly met and closed them in this release. 6 remain partial, and 1 depends on how the application is hosted.

Status	First delivery	This release	Meaning
Met	138	148	Implemented and verified in the application.
Fixed	—	10	Included in "Met": a gap found by this review and closed in this release.
Partial	16	6	Implemented in part; the remaining work is described in Section 5.
Deployment	1	1	The software provides the means; the result depends on hosting.

Note The SRS file attached for this review is identical, byte for byte, to the one used to build the application. The gaps are therefore gaps in the delivery, not changes in the specification.

1.1 What changed in this release

- **Benchmarking module (new):** compares project types, tracks and departments within an organization, and organizations within the same group. External benchmarking is out of scope.
- **Seven sectors and real groups:** Construction added; two groups (Atlas Infrastructure Holding, Crescent Agro-Energy Group) and two independent organizations.
- **10 SRS gaps closed:** per-project OBS, multilingual knowledge, per-process governance view, version comparison, lessons-learned prompt at closure, restricted evaluations, CI pipeline, backups, load capacity and touch targets.
- **Documents corrected:** the Coverage Checklist overstated SRS coverage; the User Guide covered the lifecycle only. Both are updated (Section 6).

2. Scope and Method

Each of the 155 requirements (112 functional, 43 non-functional) was checked against the application code, the automated tests and measurements taken on a freshly seeded database. The delivered documents (User Guide, Installation Guide, Coverage Checklist) were checked for what they claim and what they leave out.

Evidence type	How it was obtained
Code review	Every requirement traced to the route, library or screen that implements it.
Automated tests	11 tests: tenant isolation, RBAC and entitlement 403s, benchmarking scope, cache consistency.
Walkthrough replay	The five User Guide runs replayed through the API on a fresh database.
Measurements	Response times, 100 concurrent users, BPMN round trip, WCAG 2.1 AA scan (axe-core), phone viewport.

3. Result by SRS Section

SRS section	Req.	Met	of which fixed	Partial	Hosting
4.1 Tenant & Organizational Hierarchy	8	7	1	1	—
4.2 Identity & Role-Based Access Control	7	7	—	—	—
4.3 Configuration Management (Solution Packs, Add-Ons, Compliance Standards)	14	14	—	—	—
4.4 AI Use Case Library & Governance	11	11	—	—	—
4.5 Standards / Knowledge Base & Retrieval-Augmented Generation	4	4	1	—	—
4.6 Process/Stage Governance (Alerts, Business Rules, Controls, Risks, RACSI, Custom KPIs)	8	8	1	—	—
4.7 BPMN 2.0 Process Modeling	5	5	—	—	—
4.8 Reports & Data Export	4	4	—	—	—
4.9 Notification Center (Alerts)	5	5	—	—	—
4.10 AI Assistant / Chatbot	6	6	—	—	—
4.11 Audit Trail & Justification Governance	4	4	—	—	—
4.12 Help & In-App Guidance	3	3	—	—	—
4.13 Template Libraries	4	4	—	—	—
4.14 Multi-Language & Localization	6	5	—	1	—
4.15 Work Breakdown Structure & Gantt Scheduling	6	6	—	—	—
4.16 Version Management	5	5	1	—	—
4.17 Multi-Channel Communication	5	5	—	—	—
4.18 Return on Experience (REX)	7	7	1	—	—
5.1 Performance	4	4	—	—	—
5.2 Security	11	11	2	—	—
5.3 Reliability & Data Integrity	4	3	1	—	1
5.4 Scalability	3	2	1	1	—
5.5 Usability & Accessibility	4	3	1	1	—
5.6 Localization	1	1	—	—	—
5.7 Portability / Installability	2	1	—	1	—
5.8 Maintainability / Extensibility	5	4	—	1	—
5.9 Compatibility	3	3	—	—	—
5.10 Auditability & Compliance	4	4	—	—	—
5.11 Data Retention & Privacy	2	2	—	—	—

4. Gaps Closed in This Release

Each row was a partial requirement in the first delivery. The fix is in the application and verified as described.

Requirement	Gap and fix	Status
FR-DA-TEN-04	Added per-project OBS trees (tree selector) and free-text node types. First delivery had per-organization trees only.	Fixed
FR-DA-KB-01	Reference articles now seeded in English, French and Arabic, with a language filter. First delivery: English only.	Fixed
FR-DA-GOV-08	Macro process pages now list the organization's rules, controls, risks, RACSI activities and lessons tagged to that process. First delivery: tags stored but no per-process view.	Fixed
FR-DA-VER-03	Added "Compare two versions" field by field on every version history (registers, AI use cases, templates, REX, track configuration). First delivery: history without comparison.	Fixed
FR-DA-REX-01	Lesson prompt now also appears when a project reaches Launched (Fast Track) or Retired, and to the executives who close it. First delivery: only after task completion and Kill.	Fixed
NFR-DA-SEC-10	Individual evaluation verdicts and notes now restricted to the people involved and roles holding evaluation.view; others see "Evaluated" only. Peer organizations in benchmarks are aggregates only.	Fixed
NFR-DA-SEC-11	11 automated tests (tenant isolation and benchmarking) now run in a CI pipeline on Linux, Windows and macOS. First delivery: tests without a pipeline.	Fixed
NFR-DA-REL-04	Added automatic daily backups (VACUUM INTO) with retention, npm run backup, and a Backups panel. Off-site copy remains a hosting task.	Fixed
NFR-DA-SCALE-02	Measured 100 simultaneous users: 400 requests, 0 failures; p95 4.4 s before, 0.9 s after adding a per-user read cache.	Fixed
NFR-DA-UX-02	Verified at 390 px: no horizontal scrolling; touch targets raised to 44 px on touch screens and 24 px minimum elsewhere.	Fixed

5. Remaining Partial and Hosting Items

These items are implemented in part, or depend on hosting. Each has a concrete next step.

Requirement	Current state	Recommended next step	Status
FR-DA-TEN-08	Other tenants' records return 404 and lists are scoped (automated tests). Record ids are one global sequence, so the size of an id can hint at overall volume. Recommendation: opaque identifiers.	Replace sequential record ids in URLs and API payloads with opaque identifiers (UUID or per-tenant codes). Medium effort; touches every route.	Partial
FR-DA-I18N-01	All 1,100 interface strings and 117 fixed server messages in EN, FR, AR, plus reference labels. Remaining in English: long reference texts from the source documents (macro-process goals, 727 step descriptions) and 18 server messages that embed values.	Translate the reference catalog (macro-process goals, SIPOC texts, 727 step descriptions) with a terminology review, and move the 18 value-bearing server messages to parameterized dictionary keys.	Partial
NFR-DA-REL-02	Uptime depends on hosting. The application restarts cleanly and supports a process manager; 99.9% needs redundant hosting and monitoring.	Host behind a process manager or container orchestrator with health checks, two instances and monitoring; keep off-site backups.	Deployment
NFR-DA-SCALE-03	Data access goes through one query module and retrieval through one module, so they can be replaced; no distributed database or vector store has been plugged in to prove it.	Prove swappability with a PostgreSQL adapter for the query module and a vector-store adapter for retrieval, running the same test suite.	Partial
NFR-DA-UX-03	axe-core WCAG 2.1 AA scan of 14 screens: ARIA errors and grey-text contrast fixed. Remaining: brand-mandated white text on orange table headers (2.28:1) and orange KPI numbers (2.99:1).	Decide the brand-versus-accessibility conflict: keep the orange table header but use Grey Dark text (5.9:1), and use Grey Dark for KPI numbers or enlarge them. Both are one-line style changes.	Partial
NFR-DA-PORT-02	Verified on Linux. The CI workflow now runs install, seed, tests and build on Windows and macOS too; see its result.	Check the Windows and macOS jobs of the CI workflow after the push; fix any path or shell difference they reveal.	Partial
NFR-DA-MAINT-01	Engines are separate modules behind one REST contract; reuse by a sibling Dynamic App has not been demonstrated.	Package the task register, alert engine, RBAC, KPI and audit modules as a shared library and consume it from a second Dynamic App.	Partial

6. Gaps in the Delivered Documents

The review also compared what the documents say with what the SRS asks and what the application does.

Document	Gap found	Action taken
User Guide	Covered the project lifecycle only. Most SRS modules (governance registers, BPMN, AI use cases, knowledge, templates, WBS, REX, reports, notifications, administration, versions) had no usage scenario.	Added Part "Everyday scenarios by module" with step-by-step scenarios and the values to type, plus a full benchmarking walkthrough (Run 6).
User Guide	Did not show groups, independent organizations or the Construction sector.	Updated organization tables and added the group model to the getting-started chapter.
Coverage Checklist	Marked all 155 SRS requirements as "traced" from a group-level mapping, without checking each requirement. It overstated coverage.	Replaced by this requirement-by-requirement assessment; the checklist now links to it and shows Partial items.
Coverage Checklist / Traceability screen	The in-app traceability page labelled every requirement "Implemented".	The page now shows the assessed status (Met, Partial, Deployment) per requirement.
Installation Guide	No backup and restore instructions (NFR-DA-REL-04); six organizations only.	Added backup, restore and retention; seven organizations with their groups.
All guides	The official POWERACT logo could not be downloaded (network policy), so covers use a text wordmark.	Unchanged: place the logo at docs/brand/poweract-logo.png and rebuild.

7. Additions Made With This Review

7.1 Organizations, sectors and groups

The demonstration now covers seven sectors. An organization either belongs to a group (Group = Yes) or operates independently (Group = No), as FR-DA-TEN-02 allows.

Organization	Sector	Group	Group name
Metro City Digital Services Agency	Public Sector	No	—
Cedarline Precast Systems	Manufacturing in Construction	Yes	Atlas Infrastructure Holding
Meridale Health Network	Healthcare	No	—
Valdora Dairy Cooperative	Agro-Business – Dairy Products	Yes	Crescent Agro-Energy Group
Orvane Transit Group	Transportation	Yes	Atlas Infrastructure Holding
Kestrel Energy & Utilities	Oil, Gas & Energy	Yes	Crescent Agro-Energy Group
Ridgeway Construction Contractors	Construction	Yes	Atlas Infrastructure Holding

Every organization has 21 projects and at least 10 instances of each of the nine E2E processes (E2E-01: 21, E2E-02: 19, E2E-03: 16, E2E-04: 14, E2E-05: 16, E2E-06: 12, E2E-07: 12, E2E-08: 10, E2E-09: 13).

7.2 Benchmarking module

Scope	What is compared	Who can see it
Within the organization	Project types (Product, Service, Product-Service), tracks (Full, Light, Fast) and owning departments, with track and type filters.	Roles with benchmark.view (most internal roles).
Across the group	The organization against the other organizations of the same group, as organization-level aggregates.	Roles with benchmark.group (executives, managers, administrators).
External	Organizations outside the group, market or industry data.	Out of scope. Not available.

Thirteen indicators are compared: projects, launch rate, first-time Go rate, Recycle rate, Kill rate, gate decision time, time to market, tasks completed on time, checklist compliance, tasks rated Effective, average NPV, average ROI and lessons-learned rating. Segments with fewer than three projects are shown but not ranked.

- Only aggregates cross organization boundaries: no project, person or document of another organization is returned (automated test).
- An organization can stop sharing its aggregates with its group (benchmark.manage); peers then see "Not shared".
- An organization without a group sees no group comparison.

Example: Atlas Infrastructure Holding, seen by the Executive Sponsor of Ridgeway Construction Contractors.

Organization	First-time Go	Recycle	Time to market	Average NPV
Cedarline Precast Systems	95%	2.9%	174 days	805 kUSD
Orvane Transit Group	86.5%	11.1%	212 days	662 kUSD
Ridgeway Construction Contractors (you)	86.1%	11.4%	241 days	272 kUSD

8. Measurements

Requirement	Measure	Result	Target
NFR-DA-PERF-01	Dashboard / project list / my tasks / benchmark	31 / 27 / 2 / 170 ms	≤ 2 s
NFR-DA-PERF-02	AI suggestion, deterministic path	4 ms	≤ 5 s
NFR-DA-PERF-03	Knowledge search / help search	2 / 1 ms	≤ 1 s
NFR-DA-PERF-04	Report export PDF / Excel / Word	296 / 46 / 90 ms	≤ 3 s
NFR-DA-SCALE-02	100 simultaneous users, 400 requests	0 failures; p95 0.9 s (4.4 s before the fix)	No degradation
NFR-DA-COMPAT-02	BPMN parse, export, re-parse	63 of 63 diagrams	All
NFR-DA-UX-02	Phone width 390 px, 5 screens	No horizontal scroll; 44 px touch targets	WCAG 2.1 AA
NFR-DA-UX-03	axe-core WCAG 2.1 AA, 14 screens	2 brand-mandated contrast findings remain	No violations
NFR-DA-SEC-11	Automated tests	11 of 11 pass; CI on Linux, Windows, macOS	In pipeline

Timings are medians of five calls on the demonstration data (7 organizations, 147 projects), measured on a single server process.

Appendix. Requirement Matrix

All 155 requirements with their status in the first delivery and now. Requirement texts are shortened; the full wording is in the SRS.

ID	Requirement (shortened)	Before	Now	Evidence
FR-DA-TEN-01	The system shall support full Create, Read, Update, and Delete on Group, Organization, and Project records, gated by a...	Met	Met	Administration > Organizations & OBS: groups, organizations; Innovation projects: projects (create, edit, delete) behind hierarchy.manage / project rights.
FR-DA-TEN-02	The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully...	Met	Met	Seed: Atlas Infrastructure Holding (3 orgs), Crescent Agro-Energy Group (2 orgs), 2 independent organizations. Group is optional in the organization form.
FR-DA-TEN-03	Deleting a Group shall not delete its member Organizations; deleting an Organization shall cascade-delete its Projects and directly-scoped...	Met	Met	Foreign keys: groups ON DELETE SET NULL; organization ON DELETE CASCADE for projects, users, AI state.
FR-DA-TEN-04	The system shall support full Create, Read, Update, and Delete on Organizational Breakdown Structure (OBS) entries, each typed or...	Partial	Fixed	Added per-project OBS trees (tree selector) and free-text node types. First delivery had per-organization trees only.
FR-DA-TEN-05	The system shall prevent deleting an OBS node from breaking referential integrity, either by cascading or by re-parenting/reassigning...	Met	Met	Deleting a node moves its children up one level; linked records keep integrity (SET NULL).
FR-DA-TEN-06	The system shall display, for every OBS node, a live count of linked item types, so accountability and activity concentration are visible...	Met	Met	OBS table shows live counts of users, projects, rules, controls, risks, RACSI, BPMN and REX per node.
FR-DA-TEN-07	The system shall enforce the active Solution Pack's maximum-Projects and maximum-OBS-node quotas at creation time, server-side, rejecting...	Met	Met	checkQuota on project and OBS creation returns HTTP 409.
FR-DA-TEN-08	The system shall ensure that no Organization can enumerate, query, or infer the existence of another Organization through any API...	Partial	Partial	Other tenants' records return 404 and lists are scoped (automated tests). Record ids are one global sequence, so the size of an id can hint at overall volume....
FR-DA-RBAC-01	The system shall support full Create, Read, Update, and Delete on user accounts, including name, email/username, one or more assigned...	Met	Met	Administration > Users & roles: name, e-mail, several roles, language.
FR-DA-RBAC-02	The system shall provide a Permission Matrix screen displaying every permission/capability code against every role, editable by users...	Met	Met	Administration > Permission matrix, editable, effective immediately; new permissions added by a startup migration.
FR-DA-RBAC-03	The system shall enforce every API endpoint, server-side, against the authenticated user's resolved permission set (the union across all...	Met	Met	requirePerm on every route; 403 verified by tests.
FR-DA-RBAC-04	The system shall enforce that the Owner and the Evaluator of any two-party verification step are never the same user, both in the UI and...	Met	Met	Owner cannot evaluate own task; evaluator must differ (UI and server).
FR-DA-RBAC-05	The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assignment purposes...	Met	Met	/directory returns id, name, e-mail for assignment.
FR-DA-RBAC-06	The system shall issue a signed, time-boxed authentication token on successful login, required on every subsequent API call, and shall...	Met	Met	Signed JWT with 8-hour lifetime; password hashes never returned.
FR-DA-RBAC-07	The system shall expose a single platform-wide justification-required toggle (Section 3.7), defaulting to on, gated by a...	Met	Met	Justification-required toggle in Configuration, default on.
FR-DA-CFG-01	The system shall record, per Organization, exactly one Solution Pack as the single source of truth for that Organization's feature and...	Met	Met	One subscription per organization; entitlements.js is the single source.
FR-DA-CFG-02	The system shall define, for each Solution Pack, at minimum: which modules/features are enabled, the AI tier available (Assistive-only vs....	Met	Met	Features, AI tier and quotas (projects, OBS nodes, custom AI use cases) per pack/bundle.
FR-DA-CFG-03	The system shall record a set of priced Add-Ons, each an independent toggle that grants its associated feature to a Pack that does not...	Met	Met	28 add-ons as independent toggles, compatibility checked.
FR-DA-CFG-04	The system shall record a set of priced Compliance & Security Standards modules (at minimum GDPR, ISO/IEC 27001, and SOC 2), each an...	Met	Met	GDPR, ISO/IEC 27001, SOC 2, ISO 27701, HIPAA, ISO 9001 toggles.
FR-DA-CFG-05	The system shall, on activating a Compliance & Security Standard, idempotently seed a starting Controls scaffold tagged to that framework...	Met	Met	Idempotent starter controls; deactivation keeps tagged controls.
FR-DA-CFG-06	The system shall enforce every Pack-gated feature and every Add-On/Compliance-gated feature server-side via dedicated middleware on the...	Met	Met	requireFeatureFor middleware returns 403 naming the feature, independent of RBAC (tested).
FR-DA-CFG-07	The system shall expose the Organization's effective configuration and the static Pack/Add-On/Compliance catalog to every authenticated...	Met	Met	/config and /catalog readable by every authenticated user; menu shows locked items.
FR-DA-CFG-08	The system shall expose current usage against every quota dimension, computed live from current data, and shall visually flag a quota that...	Met	Met	Configuration shows usage bars per quota and flags reached quotas.

ID	Requirement (shortened)	Before	Now	Evidence
FR-DA-CFG-09	The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security...	Met	Met	Non-certification disclosure on activation, in Help and in the knowledge base (EN, FR, AR).
FR-DA-CFG-10	The system shall allow an authorized Organization administrator to register, configure, enable, and disable a connection to an external...	Met	Met	Administration > Integrations: register, configure, enable/disable each connector.
FR-DA-CFG-11	The system shall store every external integration's credentials using the same client-side-only or encrypted-server-side pattern already...	Met	Met	Credentials encrypted (AES-GCM) at rest, never returned or logged.
FR-DA-CFG-12	The system shall verify an external integration's connection health on demand and shall record the outcome (success, authentication...	Met	Met	Test connection records Healthy / Authentication failure / Unreachable with timestamp.
FR-DA-CFG-13	The system shall allow an authorized administrator to map fields between an external application's records and the platform's own...	Met	Met	Field mappings and exchange log (record, time, direction, result).
FR-DA-CFG-14	The system shall accept inbound events from an external application only through the Multi-Channel Communication webhook channel (Section...	Met	Met	Inbound endpoint verifies HMAC signature and scopes events to the registering organization.
FR-DA-AI-01	The system shall support full Create, Read, Update, Delete, and version-revert on an AI Use Case catalog, each entry capturing an id,...	Met	Met	AI use cases: create, edit, versions, revert; ai.manage.
FR-DA-AI-02	The system shall seed every Organization with the same library of AI Use Cases on creation, and shall allow an Organization to...	Met	Met	23 use cases seeded per organization; custom use cases limited by quota.
FR-DA-AI-03	The system shall create Version 1 of an AI Use Case automatically on creation, mark it Default/Current, and shall retain the full version...	Met	Met	Version 1 on creation; revert creates a new current version.
FR-DA-AI-04	The system shall allow an Organization Administrator to activate or deactivate any AI Use Case for the whole Organization, and shall allow...	Met	Met	Organization activation plus project Inherit/On/Off override.
FR-DA-AI-05	The system shall block, server-side, any new AI suggestion from a deactivated AI Use Case (by its effective state) immediately on...	Met	Met	Generation refused for a deactivated use case; reviewed content kept.
FR-DA-AI-06	The system shall visually label every AI-generated output, wherever shown, as AI-generated and requiring human review, alongside its tier...	Met	Met	Every suggestion labelled AI-generated with tier and source.
FR-DA-AI-07	The system shall write an append-only AI Usage Log entry for every AI suggestion's outcome (Accepted, Edited, or Rejected), capturing the...	Met	Met	Append-only usage log enforced by database triggers.
FR-DA-AI-08	The system shall allow a user to configure a Real LLM Provider Connection (provider, model, endpoint, API key) that is persisted...	Met	Met	Settings > Live AI model stored in the browser only, forwarded per request.
FR-DA-AI-09	The system shall, on any failure of a live LLM call, fall back automatically to the AI Use Case's deterministic built-in generator, so a...	Met	Met	Live call failure falls back to the built-in generator.
FR-DA-AI-10	The system shall ensure an AI Use Case's activation, tier, or generation never grants a user visibility into data beyond what their RBAC...	Met	Met	Suggestions read only records the user may already see.
FR-DA-AI-11	The system shall, before generating any suggestion, ground it in retrieved reference content (Section 4.5), instruct the underlying model...	Met	Met	Retrieval grounding; references shown with the output.
FR-DA-KB-01	The system shall maintain a Standards/Knowledge Base of reference content — regulatory or methodological standards, or...	Partial	Fixed	Reference articles now seeded in English, French and Arabic, with a language filter. First delivery: English only.
FR-DA-KB-02	The system shall index this knowledge base, the tenant's own historical records, and a static application-feature/FAQ corpus using the...	Met	Met	One TF-IDF engine for knowledge, records, FAQ and assistant help mode.
FR-DA-KB-03	The system shall scope every retrieval query to the requesting user's own tenant; no cross-tenant record shall appear in a retrieval...	Met	Met	Retrieval scoped to the tenant (tested).
FR-DA-KB-04	The system shall gracefully degrade when a live LLM call is unavailable or fails: the retrieval result (the matched reference content, the...	Met	Met	Retrieval result returned when the live model is unavailable.
FR-DA-GOV-01	The system shall support full Create, Read, Update, and Delete on Business Rules, each with a condition, an action, a severity, and an...	Met	Met	Business rules register with condition, action, severity, owner.
FR-DA-GOV-02	The system shall support full Create, Read, Update, and Delete on Controls, each classified under one of the five COSO Internal Control —...	Met	Met	Controls with COSO component, type, frequency, owner, effectiveness.
FR-DA-GOV-03	The system shall allow a Control to be optionally tagged with the Compliance & Security Standard it was seeded under or otherwise...	Met	Met	Compliance standard tag on controls, with non-certification wording.
FR-DA-GOV-04	The system shall support full Create, Read, Update, and Delete on Risk/Opportunity items, each with a type, a likelihood (1–5) and impact...	Met	Met	Risks and opportunities, computed score, 5×5 heat map.
FR-DA-GOV-05	The system shall support full Create, Read, Update, and Delete on Custom KPIs, additive to the platform's built-in KPI set, each with a...	Met	Met	Custom KPIs with formula, target, owner, measurements.
FR-DA-GOV-06	The system shall support a RACSI accountability matrix: full Create, Read, Update, and Delete on RACSI Activities, each optionally linked...	Met	Met	RACSI activities linked to steps or records; assignments to people or roles.

ID	Requirement (shortened)	Before	Now	Evidence
FR-DA-GOV-07	The system shall enforce that a RACSI Activity has at most one Accountable assignment at any time, both in the UI and at the...	Met	Met	One Accountable per activity enforced by a unique index.
FR-DA-GOV-08	The system shall allow a Business Rule, a Control, a Risk/Opportunity, and a RACSI Activity to each be optionally tagged with the owning...	Partial	Fixed	Macro process pages now list the organization's rules, controls, risks, RACSI activities and lessons tagged to that process. First delivery: tags stored but no...
FR-DA-BPMN-01	The system shall support full Create, Read, Update, and Delete on BPMN Diagrams, each with a title, description, and a BPMN 2.0-compliant...	Met	Met	bpmn-js modeler with XML payload, title and description.
FR-DA-BPMN-02	The system shall grant read-only pan/zoom viewing of a BPMN Diagram to any user holding view access, and shall grant the full editing...	Met	Met	Viewer for view rights; modeler and Save for edit rights with the entitlement.
FR-DA-BPMN-03	The system shall render the BPMN editor's tool palette in a dedicated panel beside the diagram canvas, structurally separate from the...	Met	Met	Palette in its own panel beside the canvas.
FR-DA-BPMN-04	The system shall provide Export (download the current diagram as a schema-valid .bpmn/XML file) and Import (load a .bpmn/XML file,...	Met	Met	Export .bpmn and Import .bpmn.
FR-DA-BPMN-05	The system shall be seeded, per Organization, with at least one BPMN Diagram modeling the application's own core process end-to-end.	Met	Met	9 diagrams per organization (one per E2E process).
FR-DA-REP-01	The system shall compute the platform's standard KPIs in real time from current data, scoped to the requesting user's tenant.	Met	Met	38 KPIs computed live from tenant data.
FR-DA-REP-02	The system shall provide a set of standard reports, each exportable on demand as a downloadable PDF, Excel (.xlsx), and Word (.docx) file,...	Met	Met	24 reports exported server-side as PDF, Excel and Word in the brand style.
FR-DA-REP-03	The system shall name every exported report file with the report key and the export date, and shall serve it with the correct MIME type...	Met	Met	File name with report key and date; correct MIME type and attachment header.
FR-DA-REP-04	The system shall offer a client-side CSV export on its highest-traffic tables, reflecting exactly the rows and columns currently on screen...	Met	Met	CSV of on-screen rows with UTF-8 byte-order mark on every table.
FR-DA-ALT-01	The system shall define a shared Alert Catalog naming every alert type once, with a human-readable name and a primary process/stage tag,...	Met	Met	One shared alert catalog (31 types) for seed and live checks.
FR-DA-ALT-02	The system shall surface a persistent, dismissible per-tenant or per-project alert inbox, with a live unread-alert indicator in the...	Met	Met	Alert inbox with unread counter in the top bar.
FR-DA-ALT-03	The system shall avoid raising a duplicate alert of the same type against the same triggering entity within the same period.	Met	Met	De-duplication by type, entity and period.
FR-DA-ALT-04	The system shall allow a user to mark an individual alert as read, and, where dismissal is supported, shall persist the dismissal...	Met	Met	Read and dismiss stored per user, separate from the alert.
FR-DA-ALT-05	The system shall allow an authorized user to independently enable or disable each alert type, displaying its human-readable name alongside...	Met	Met	Alert types enabled/disabled per organization, name and code shown.
FR-DA-AST-01	The system shall provide a data-query mode that answers natural-language questions about the requesting user's own organization by...	Met	Met	Fixed data-query intents answered in EN, FR and AR.
FR-DA-AST-02	The system shall associate every data-query intent with the single RBAC permission required to answer it, and shall evaluate that...	Met	Met	Permission checked after matching, before querying; refusal names the permission.
FR-DA-AST-03	The system shall scope every data-query answer to the requesting user's own tenant, consistent with the platform's multi-tenancy...	Met	Met	Answers scoped to the tenant.
FR-DA-AST-04	The system shall provide an application-help mode answering "Can it...?" / "How do I...?" questions by ranking a static FAQ/feature corpus...	Met	Met	Help mode ranks the FAQ corpus with the shared engine.
FR-DA-AST-05	The system shall render the AI Assistant both as a dedicated screen and as a floating widget accessible from every authenticated screen,...	Met	Met	Dedicated screen and floating widget on every screen.
FR-DA-AST-06	The system shall gate the AI Assistant behind a single view permission and the active Solution Pack's AI Assistant entitlement,...	Met	Met	assistant.use permission plus assistant entitlement.
FR-DA-AUD-01	The system shall write an audit or change-log entry for every state-changing operation on a governed field, capturing the actor,...	Met	Met	Audit entry per governed change with actor, time, entity, before/after.
FR-DA-AUD-02	The system shall follow the stage-then-justify pattern (Section 3.7) for every score or state change to a governed record: staging a new...	Met	Met	Stage then justify; value and note written together.
FR-DA-AUD-03	The system shall make the audit/change log viewable, read-only, to users holding an audit-view permission, and shall never expose an edit...	Met	Met	Read-only audit trail screen; no edit or delete route.
FR-DA-AUD-04	The system shall ensure audit-log writes never block or fail the primary operation they accompany.	Met	Met	Audit writes wrapped so they never fail the operation.
FR-DA-HLP-01	The system shall provide an in-app Help screen presenting topic articles covering every module of the application, accessible to every...	Met	Met	24 Help articles covering every module, open to every user.

ID	Requirement (shortened)	Before	Now	Evidence
FR-DA-HLP-02	The system shall render Help content in the user's currently selected display language, matching the article set one-to-one across every...	Met	Met	Same article set in EN, FR and AR.
FR-DA-HLP-03	The system shall provide a free-text search box on the Help screen filtering topics by a case-insensitive match against the topic title or...	Met	Met	Case-insensitive search on title and body.
FR-DA-TPL-01	The system shall support a library of pre-filled starting points (templates) for its principal record type, each capturing default field...	Met	Met	Project templates pre-fill the New project wizard; every field stays editable.
FR-DA-TPL-02	The system shall support full Create, Read, Update, and Delete on templates, versioned via the same version-history mechanism as the AI...	Met	Met	Templates CRUD with versions and restore.
FR-DA-TPL-03	The system shall make templates available regardless of Solution Pack, consistent with the always-standard treatment of Custom KPIs and...	Met	Met	Templates available in every subscription.
FR-DA-TPL-04	Deleting a template shall not affect any record already created from it; a template is a starting point copied at use time, not a...	Met	Met	Templates copied at use; deleting one leaves projects untouched.
FR-DA-I18N-01	The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and...	Partial	Partial	All 1,100 interface strings and 117 fixed server messages in EN, FR, AR, plus reference labels. Remaining in English: long reference texts from the source documents...
FR-DA-I18N-02	The system shall allow a user to change their display language at any time without reloading or losing unsaved work.	Met	Met	Language switch in the top bar applies at once without reload.
FR-DA-I18N-03	The system shall persist a user's language preference and apply it automatically on subsequent logins, resolved per the precedence order...	Met	Met	Language saved per user; user > organization > English.
FR-DA-I18N-04	The system shall render the entire application right-to-left, including navigation, forms, tables, and charts, when Arabic is selected.	Met	Met	Arabic mirrors navigation, forms, tables, charts and the Gantt.
FR-DA-I18N-05	The system shall allow an Organization to configure a tenant-level default language.	Met	Met	Organization default language in Configuration.
FR-DA-I18N-06	The system shall allow adding a new supported language by configuration — populating the shared translation dictionary (Section 3.6) —...	Met	Met	A new language is a new JSON file in server/src/i18n.
FR-DA-WBS-01	The system shall allow any set of Actions the user selects to be grouped into a Work Breakdown Structure (WBS), organized as a tree of WBS...	Met	Met	WBS trees referencing tasks (actions) from any project.
FR-DA-WBS-02	The system shall allow a WBS node or a referenced Action to carry a planned start date, planned end date or duration, percent-complete,...	Met	Met	Dates, duration, percent complete, predecessors.
FR-DA-WBS-03	The system shall render a WBS as a Gantt chart, plotting every node and its referenced Actions on a shared timeline, drawing dependency...	Met	Met	Gantt with dependency lines and state colours.
FR-DA-WBS-04	The system shall recompute a WBS node's own planned dates and percent-complete from its child nodes/Actions when the node itself is not...	Met	Met	Summary nodes roll up dates and progress.
FR-DA-WBS-05	The system shall support full Create, Read, Update, and Delete on WBS structures, and shall allow the same Action to be referenced from...	Met	Met	WBS CRUD; a task can appear in several WBS.
FR-DA-WBS-06	The system shall export a WBS/Gantt view to at least one portable format (e.g. PDF or an image), consistent with the platform's standard...	Met	Met	Gantt export to PDF.
FR-DA-VER-01	The system shall provide a single, generic Version Management service (Section 3.8) that any governed entity type can register with,...	Met	Met	One version service (entity_versions) used by all versioned entities.
FR-DA-VER-02	The system shall create an initial version automatically when a versioned entity is created, mark it Current, and create a new version —...	Met	Met	Version 1 at creation, new version on each edit.
FR-DA-VER-03	The system shall retain every version of a versioned entity indefinitely, with no hard-delete route for a historical version, and shall...	Partial	Fixed	Added "Compare two versions" field by field on every version history (registers, AI use cases, templates, REX, track configuration). First delivery: history without...
FR-DA-VER-04	The system shall allow an authorized user to revert a versioned entity to any historical version; reverting shall duplicate that version's...	Met	Met	Restore creates a new current version.
FR-DA-VER-05	The system shall record, on every version, the acting user, a timestamp, and — where the platform-wide justification toggle (Section 3.7)...	Met	Met	User, time and justification stored on each version.
FR-DA-COMM-01	The system shall support delivering a notification or alert (Section 4.9) through one or more channels — at minimum in-app, email, and...	Met	Met	One dispatch interface: in-app, e-mail, webhook, SMS, push.
FR-DA-COMM-02	The system shall allow a user to independently enable or disable each available channel for each notification category, defaulting every...	Met	Met	Preferences per category and channel, in-app on by default.
FR-DA-COMM-03	The system shall compose a message once, per notification event, and render it in the recipient's own display language (Section 3.6)...	Met	Met	Message composed once in the recipient's language.
FR-DA-COMM-04	The system shall record a delivery-status record per message per channel (queued, sent, delivered,	Met	Met	Delivery status per channel with independent retries.

ID	Requirement (shortened)	Before	Now	Evidence
	failed), and shall retry a failed...			
FR-DA-COMM-05	The system shall never send a tenant's data through an external channel provider (email, SMS, push) beyond the minimum content required...	Met	Met	Minimal content per channel; tenant-scoped dispatch.
FR-DA-REX-01	The system shall prompt for a REX entry when the application's principal record reaches a terminal/closed status, and when an individual...	Partial	Fixed	Lesson prompt now also appears when a project reaches Launched (Fast Track) or Retired, and to the executives who close it. First delivery: only after task completion...
FR-DA-REX-02	The system shall structure every REX entry with, at minimum: what went well, what did not, a root-cause note, a recommendation/improvement...	Met	Met	Went well, went wrong, root cause, recommendation, category, rating 1-5.
FR-DA-REX-03	The system shall support full Create, Read, Update, and Delete on REX entries, each optionally tagged to the owning OBS node and to the...	Met	Met	REX CRUD with OBS and process tags.
FR-DA-REX-04	The system shall index every saved REX entry into the same retrieval corpus used for AI Use Case grounding, the AI Assistant, and...	Met	Met	REX indexed in the shared retrieval corpus.
FR-DA-REX-05	The system shall provide a REX Register view aggregating entries by category/tag and by rating, so a recurring root cause or a declining...	Met	Met	REX register by category and rating trend.
FR-DA-REX-06	The system shall version every REX entry via the generic Version Management service (Section 3.8/4.16), retaining its full edit history...	Met	Met	REX versioned with restore.
FR-DA-REX-07	The system shall gate REX entry management behind a dedicated capability, and shall scope every REX read and every retrieval-grounded...	Met	Met	rex.manage capability; tenant-scoped reads.
NFR-DA-PERF-01	Dashboard and list views shall render within 2 seconds under nominal per-tenant data volumes.	Met	Met	Measured: dashboard 31 ms, project list 27 ms, my tasks 2 ms, benchmark 170 ms (target 2 s).
NFR-DA-PERF-02	An AI agent suggestion shall return within 5 seconds under its deterministic path.	Met	Met	Measured: deterministic suggestion 4 ms (target 5 s).
NFR-DA-PERF-03	A knowledge-base or FAQ retrieval search shall return ranked results within 1 second for a tenant's full corpus.	Met	Met	Measured: knowledge search 2 ms, help search 1 ms (target 1 s).
NFR-DA-PERF-04	A single standard-report export (PDF, Excel, or Word) shall complete within 3 seconds for a tenant's current data volume.	Met	Met	Measured: PDF 296 ms, Excel 46 ms, Word 90 ms (target 3 s).
NFR-DA-SEC-01	All API endpoints except login shall require a valid, signed, time-boxed authentication token.	Met	Met	All routes except login and dictionaries require a token (tested).
NFR-DA-SEC-02	Passwords shall be stored only as salted cryptographic hashes; plaintext passwords shall never be logged or persisted.	Met	Met	bcrypt hashes; passwords never logged.
NFR-DA-SEC-03	Every authorization decision shall be enforced server-side; client-side permission and Pack-entitlement checks are a UX convenience only,...	Met	Met	Server-side checks; client checks are UX only.
NFR-DA-SEC-04	No data query shall be constructed by concatenating untrusted input; all data access shall use parameterized statements or an equivalent...	Met	Met	Parameterized statements throughout.
NFR-DA-SEC-05	Owner/Evaluator segregation of duties, and any equivalent server-enforced business rule, shall be enforced server-side, not merely hidden...	Met	Met	Segregation of duties and CTL-01 enforced by the server.
NFR-DA-SEC-06	A one-Accountable-per-RACSI-activity rule, and any equivalent uniqueness rule, shall be enforced by a data-persistence-layer constraint,...	Met	Met	Unique index for one Accountable per activity.
NFR-DA-SEC-07	A Real LLM Provider Connection's API key shall be captured and stored only client-side; the backend shall never persist it to any table,...	Met	Met	API key kept client-side; used for one outbound call.
NFR-DA-SEC-08	Every quota check (Projects, OBS nodes, AI Use Cases, and any other Pack-bound dimension) shall be enforced at the point of...	Met	Met	Quotas checked at creation; 409.
NFR-DA-SEC-09	Solution Pack entitlement shall be treated as an authorization axis independent of, and enforced in addition to, RBAC permissions; a route...	Met	Met	Entitlement and RBAC checked independently (tested).
NFR-DA-SEC-10	Individually identifiable records requiring restricted visibility (e.g. individual-level readiness or sentiment data) shall be restricted,...	Partial	Fixed	Individual evaluation verdicts and notes now restricted to the people involved and roles holding evaluation.view; others see "Evaluated" only. Peer organizations in...
NFR-DA-SEC-11	Tenant data isolation shall be verified by an automated cross-tenant access test suite exercised in the build/release pipeline, not solely...	Partial	Fixed	11 automated tests (tenant isolation and benchmarking) now run in a CI pipeline on Linux, Windows and macOS. First delivery: tests without a pipeline.
NFR-DA-REL-01	The system shall degrade gracefully if an AI agent or live-LLM call fails: the underlying CRUD operation shall still succeed, with the AI...	Met	Met	AI failure never blocks the operation; fallback served.
NFR-DA-REL-02	The system shall target 99.9% uptime in a production deployment.	Deployment	Deployment	Uptime depends on hosting. The application restarts cleanly and supports a process manager; 99.9% needs redundant hosting and monitoring.
NFR-DA-REL-03	A report-export failure shall return a structured error without corrupting or partially writing the underlying report data view.	Met	Met	Export errors return a structured message; data untouched.
NFR-DA-REL-04	The persistence layer shall support scheduled backup with a defined retention policy; a production	Partial	Fixed	Added automatic daily backups (VACUUM INTO) with retention, npm run backup, and a

ID	Requirement (shortened)	Before	Now	Evidence
	deployment shall not rely on a single...			Backups panel. Off-site copy remains a hosting task.
NFR-DA-SCALE-01	The architecture shall support horizontal scaling of the web tier independently of the API tier.	Met	Met	Web built as static files, served independently from the API.
NFR-DA-SCALE-02	The system shall support at least 100 concurrent users per tenant without functional degradation.	Partial	Fixed	Measured 100 simultaneous users: 400 requests, 0 failures; p95 4.4 s before, 0.9 s after adding a per-user read cache.
NFR-DA-SCALE-03	The persistence schema and the retrieval index shall each be swappable for a distributed equivalent (e.g. a managed relational database, a...	Partial	Partial	Data access goes through one query module and retrieval through one module, so they can be replaced; no distributed database or vector store has been plugged in to...
NFR-DA-UX-01	The UI, and every generated document, shall follow the suite's common visual identity (color palette, typography, iconography)...	Met	Met	Shared tokens for UI, reports and documents.
NFR-DA-UX-02	The UI shall be responsive and usable on desktop, tablet, and mobile viewport widths, with touch-target sizing meeting WCAG 2.1 AA on...	Partial	Fixed	Verified at 390 px: no horizontal scrolling; touch targets raised to 44 px on touch screens and 24 px minimum elsewhere.
NFR-DA-UX-03	All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).	Partial	Partial	axe-core WCAG 2.1 AA scan of 14 screens: ARIA errors and grey-text contrast fixed. Remaining: brand-mandated white text on orange table headers (2.28:1) and orange KPI...
NFR-DA-UX-04	Status, severity, and classification fields shall use one consistent color/badge vocabulary across every module, so a user who has learned...	Met	Met	One status badge vocabulary across modules.
NFR-DA-I18N-01	Every layout — navigation, tables, and charts — shall render correctly mirrored under Arabic RTL, not just with mirrored text direction.	Met	Met	Layouts mirrored in Arabic, verified by screenshots.
NFR-DA-PORT-01	Installation shall succeed on a fresh target machine with only a supported Node.js runtime installed — no native C/C++ build toolchain, no...	Met	Met	Fresh install from the zip needs only Node.js 22.13+ (verified).
NFR-DA-PORT-02	The same source shall run unmodified across supported operating systems via standard package-manager install/build/start commands.	Partial	Partial	Verified on Linux. The CI workflow now runs install, seed, tests and build on Windows and macOS too; see its result.
NFR-DA-MAINT-01	Cross-cutting engines — the Action/Task Register, the Alert & Notification Engine, the RBAC layer, the KPI/Reporting engine, and the Audit...	Partial	Partial	Engines are separate modules behind one REST contract; reuse by a sibling Dynamic App has not been demonstrated.
NFR-DA-MAINT-02	AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an...	Met	Met	AI generation behind one interface (deterministic or live).
NFR-DA-MAINT-03	The Solution Pack feature and quota matrix shall be defined in exactly one place and consumed by every enforcing route, so entitlements...	Met	Met	Entitlements in one module.
NFR-DA-MAINT-04	The AI Use Case Library's generation service shall be the single designated server-side integration point for a live LLM call and its...	Met	Met	Only ai.js calls a live model.
NFR-DA-MAINT-05	The alert type-to-name and type-to-stage/process mapping shall be defined in exactly one shared module, consumed by both seed data and...	Met	Met	Alert catalog in one module.
NFR-DA-COMPAT-01	The application shall target evergreen desktop, tablet, and mobile browsers; no legacy-browser support is in scope.	Met	Met	Evergreen browsers; checked in Chromium at desktop and phone widths.
NFR-DA-COMPAT-02	BPMN diagrams shall be stored and exchanged as schema-valid BPMN 2.0 XML, verified by a round-trip import/export, so a diagram authored in...	Met	Met	Round trip verified: 63 of 63 stored diagrams parse, export and re-parse with bpmn-moddle.
NFR-DA-COMPAT-03	The application's core workflows (record CRUD, approvals, alerts) shall remain fully usable on a mobile viewport without requiring a...	Met	Met	Core workflows usable on a phone; same REST API.
NFR-DA-COMP-01	Every AI-generated suggestion shall be logged with a confidence indicator and be attributable to a specific agent/use case and record, or...	Met	Met	Usage log with confidence, use case, record or question.
NFR-DA-COMP-02	The Controls module shall support demonstrating COSO Internal Control — Integrated Framework coverage across all five components.	Met	Met	COSO coverage chart across five components.
NFR-DA-COMP-03	The system shall retain the full version history of every AI Use Case and every template indefinitely (no hard deletion of prior versions...	Met	Met	AI use case and template versions kept indefinitely.
NFR-DA-COMP-04	The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security...	Met	Met	No certification claim in UI, Help, knowledge or documents.
NFR-DA-DATA-01	By default, and whenever no Real LLM Provider Connection is configured, the system shall not transmit tenant data to any third-party...	Met	Met	No third-party call unless a user configures a live model; never for data queries.
NFR-DA-DATA-02	The system shall support both shared-infrastructure (SaaS) and dedicated/sovereign deployment options per Organization, without changing...	Met	Met	Deployment option (SaaS shared or dedicated/sovereign) recorded per organization; same isolation code.